MMET 383 · LAB 11

# Trivia Game

## Using ADO.NET data binding in Visual Basic

Step by step, with every line of code

# What we will build

- The form shows one question and four choices: A, B, C, D
- The user picks one choice and clicks **Submit Answer**
- Wrong answer: a message shows the correct letter
- Then the next question appears
- After the last question: show the number of wrong answers

# Four database words

| Word | Meaning | In this lab |
|---|---|---|
| Database | A data file (like an Excel workbook) | Trivia2003.mdb |
| Table | One sheet inside the file | Game1 |
| Record | One row | One question |
| Field | One column | Question, A, B, C, D, CorrectAnswer |

CorrectAnswer holds one letter: A, B, C, or D.

# ADO.NET: the four parts Visual Studio makes

| Part | What it does | Think of it as |
|---|---|---|
| TriviaDataSet | A copy of the data in memory | A photocopy |
| Game1TableAdapter | Moves data from the file into the DataSet (Fill) | A mover |
| Game1BindingSource | Points to the current record | A bookmark |
| Binding | Links a TextBox to one field | Bookmark moves, text changes |

**Key idea:** call MoveNext(), the bookmark moves one row, and every bound TextBox updates by itself.

# Step 1: Create the project and connect the database

1. New project: **Windows Forms App (.NET Framework)**, Visual Basic
2. View → Other Windows → Data Sources → Add New Data Source
3. Database → Dataset → New Connection
4. Microsoft Access Database File → Browse to the .mdb → Test Connection
5. "Copy the file to your project?" → Yes
6. Check Tables → Game1 → Finish

Visual Studio 2022 may say "Jet.OLEDB.4.0 ... 32-bit only". This affects the designer only. Press F5: if the questions show, the program works.

# Step 2: Build the form

| Control | Name | Text or setting |
|---|---|---|
| Label | Label1 | Question: |
| TextBox | QuestionTextBox | Multiline, ReadOnly |
| TextBox × 4 | ATextBox ... DTextBox | ReadOnly |
| RadioButton × 4 | RadioButton1 ... RadioButton4 | A, B, C, D |
| Button | Button1 | Submit Answer |
| Button | Button2 | Exit |

The code uses these names. Keep them the same.

# Step 3: Bind the fields

1. Data Sources window → expand Game1
2. Drag Question, A, B, C, D onto the form
3. Do **not** drag CorrectAnswer (it would show the answer)
4. Delete the navigator bar (◀ ▶) at the top of the form
5. Set every TextBox: ReadOnly = True

Backup way: TextBox → Properties → (DataBindings) → Text → Game1BindingSource → field

# Step 4: What the Submit button does

1. Read the user's choice
2. Find the correct answer for this question
3. Compare them and count mistakes
4. Go to the next question, or end the game

The next 6 slides are the whole Form1.vb, in order. Type them top to bottom.

# Code 1 of 6: Class and variables

The top of Form1.vb

```
Public Class Form1

    ' Form-level variables keep their values between button clicks
    Private Total As Integer       ' number of questions
    Private Incorrect As Integer   ' number of wrong answers
```

# Code 2 of 6: Form Load

Runs once, when the program starts

```
Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load
    ' Copy every row of Game1 from the database into memory
    Me.Game1TableAdapter.Fill(Me.Trivia2003DataSet.Game1)

    Total = Trivia2003DataSet.Game1.Count   ' how many questions
    Incorrect = 0                           ' no mistakes yet
    ClearSelection()                        ' no answer selected
    ShowQuestionNumber()                    ' "Question 1 of N:"
End Sub
```

# Code 3 of 6: Submit, part 1

Read the user's choice

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    ' Step 1: read the user's choice (starts empty on every click)
    Dim UserAnswer As String = ""

    Select Case True
        Case RadioButton1.Checked
            UserAnswer = "A"
        Case RadioButton2.Checked
            UserAnswer = "B"
        Case RadioButton3.Checked
            UserAnswer = "C"
        Case RadioButton4.Checked
            UserAnswer = "D"
    End Select

    ' Nothing selected: warn the user and stop here
    If UserAnswer = "" Then
        MsgBox("Please select an answer.", MsgBoxStyle.Exclamation, "Trivia Game Project")
        Exit Sub
    End If
```

# Code 4 of 6: Submit, part 2

Find the correct answer and check it

```
        ' Step 2: find the correct answer for the current question
        ' Position = index of the current record (first question = 0)
        Dim record As Integer = Game1BindingSource.Position
        Dim AnswerKey As String = Trivia2003DataSet.Game1(record).CorrectAnswer.Trim().ToUpper()

        ' Step 3: compare and count mistakes
        If UserAnswer <> AnswerKey Then
            Incorrect += 1
            MsgBox("Incorrect! The answer is " & AnswerKey, MsgBoxStyle.Information, "Trivia Game Project")
        End If
```

# Code 5 of 6: Submit, part 3

Next question, or finish the game

```
        ' Step 4: go to the next question, or end the game
        ' The last question has index Total - 1
        If record < Total - 1 Then
            Game1BindingSource.MoveNext()   ' every bound TextBox updates
            ClearSelection()
            ShowQuestionNumber()
        Else
            MsgBox("Number of incorrect is: " & CStr(Incorrect), MsgBoxStyle.Information, "Trivia Game Project")
            Button1.Enabled = False         ' game over
        End If
    End Sub
```

# Code 6 of 6: Exit and helpers

The end of Form1.vb

```
    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        Me.Close()    ' Exit: close the form normally
    End Sub

    ' Helper: uncheck all four choices
    Private Sub ClearSelection()
        RadioButton1.Checked = False
        RadioButton2.Checked = False
        RadioButton3.Checked = False
        RadioButton4.Checked = False
    End Sub

    ' Helper: show "Question X of N:" (Position starts at 0, so add 1)
    Private Sub ShowQuestionNumber()
        Label1.Text = "Question " & CStr(Game1BindingSource.Position + 1) & " of " & CStr(Total) & ":"
    End Sub

End Class
```